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CANOPY FOR ATTACHMENT TO HELICOPTER ROTOR BLADES

Abstract

A canopy apparatus for protecting individuals from weather elements near a helicopter includes a pair of rotor blade pockets which mount to adjacent rotor blades of the helicopter at their free ends. A flexible sheet extends between the rotor blade pockets and toward a rotor head of the helicopter. A rotor head pocket is attached to the flexible sheet and is shaped to receive the rotor head of the helicopter. Additional straps are used to further secure the flexible sheet to the rotor blades. The canopy apparatus mounts to the helicopter such that the rotor blades may be rotated to adjust a position of the canopy apparatus with respect to a body of the helicopter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] Not Applicable
STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
[0002] Not Applicable
THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT
[0003] Not Applicable
INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM
[0004] Not Applicable
STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR
[0005] Not Applicable
BACKGROUND OF THE INVENTION
(1) Field of the Invention
[0006] The disclosure relates to canopies and more particularly pertains to a new canopy for protecting individuals from weather elements near a helicopter.
(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98
[0007] The prior art discloses canopies which mount to helicopter rotor blades in order to provide shade or protection from precipitation to mechanics or other individuals near a helicopter. Some canopies known to the prior art include pockets which mount to the free ends of rotor blades of the helicopter. However, the prior art fails to describe such a canopy which uses a rotor head pocket to receive a rotor head of the helicopter and facilitate repositioning of the canopy by rotating the rotor blades with respect to a body of the helicopter.
BRIEF SUMMARY OF THE INVENTION
[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a pair of rotor blade pockets, each of which is shaped to receive therein a free end of a corresponding rotor blade of a helicopter therein. A flexible sheet is coupled to and extends between the pair of rotor blade pockets. The flexible sheet has a top side and a bottom side. Each rotor blade pocket is positioned on a corresponding mounting edge of a pair of mounting edges of the flexible sheet. Each mounting edge of the flexible sheet extends from the corresponding rotor blade pocket toward a vertex. A rotor head pocket is coupled to the flexible sheet at the vertex. The rotor head pocket has an opening extending therein which faces downwardly with respect to the flexible sheet during normal use. The rotor head pocket is shaped to receive a rotor head of the helicopter therein.
[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.
[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)
[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. **1** is a perspective in-use view of a canopy apparatus according to an embodiment of the disclosure.
[0013] FIG. **2** is a top in-use view of an embodiment of the disclosure.
[0014] FIG. **3** is a detail view of an embodiment of the disclosure taken from Window **3** in FIG. **2**.
[0015] FIG. **4** is a cross-section view of an embodiment of the disclosure taken from Line **4-4** in FIG. **3**.
[0016] FIG. **5** is a top view of an embodiment of the disclosure in a stored configuration.
[0017] FIG. **6** is a bottom view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0018] With reference now to the drawings, and in particular to FIGS. **1** through **6** thereof, a new canopy embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0019] As best illustrated in FIGS. **1** through **6**, the canopy apparatus **10** generally comprises a pair of rotor blade pockets **12**, each of which is shaped to receive a free end **74** of a rotor blade **72** of a helicopter **70** therein. A flexible sheet **16** is coupled to and extends between the pair of rotor blade pockets **12**. The flexible sheet **16** has a triangular shape and has a top side **18** and a bottom side **20**. The flexible sheet **16** has a pair of mounting edges **22** which converge toward a vertex **26**. Each rotor blade pocket **12** is positioned at a distal end **24** of a respective one of the mounting edges **22** relative to the vertex **26**. Each rotor blade pocket **12** has an open end **14** which faces generally toward the vertex **26** during normal use.

[0020] A rotor head pocket **28** is coupled to the flexible sheet **16** at the vertex **26**. The rotor head pocket **28** has an opening **30** which extends therein and which faces downwardly with respect to the flexible sheet **16** during normal use. The rotor head pocket **28** is shaped and configured to receive a rotor head **76** of the helicopter **70** therein. The rotor head pocket **28** has a cylindrical shape with a central axis that extends orthogonally through the opening **30**.

[0021] A plurality of mounting straps **32** is coupled to the flexible sheet **16**. Each mounting strap **32** of the plurality of mounting straps **32** is mounted to the flexible sheet **16** on an associated mounting edge **22** of the pair of mounting edges **22**. Each mounting strap **32** of the plurality of mounting straps **32** is sized to surround a corresponding one of the pair of rotor blades **72**. Each mounting strap **32** comprises a pair of segments **34** and a connector **38**. Each segment **34** of each mounting strap **32** is coupled to and extends away from the flexible sheet **16**.

[0022] The connector **38** of each mounting strap **32** is positioned on free ends **36** of the segments **34** to selectively attach the free ends **36** together, forming a closed loop to surround a corresponding rotor blade **72** of the helicopter **70**. The connector **38** of each mounting strap **32** comprises a parachute buckle but may comprise a hook-and-loop fastener, a snap fastener, or any other suitable fastening means. In other embodiments, the segments **34** of each mounting strap **32** may be tied together to mount the canopy apparatus **10** to the rotor blades **72**. Each mounting strap **32** is adjustable in length such that the mounting strap **32** is cinchable around the associated rotor blade **72**. At least one of the segments **34** of each mounting strap **32** is slidable through the connector **38** to facilitate this cinching action.

[0023] The canopy apparatus **10** may be positioned in a stored configuration **40** in which the flexible sheet **16**, the rotor blade pockets **12**, and the mounting straps **32** are positioned in the rotor head pocket **28**. A carrying strap **42** is attached to the rotor head pocket **28** to facilitate carrying the canopy apparatus **10** in the stored configuration **40**. In some embodiments the rotor head pocket **28** may be removably attached to the flexible sheet **16** via one or more fasteners. The fasteners may comprise additional parachute buckles or other suitable fastening means. The fasteners may also comprise connecting straps which are adjustable in length. The flexible sheet **16**, the mounting straps **32**, and one of the rotor blade pockets **12** also may be packed into another one of the rotor blade pockets **12**, resulting in a bundle that is storable in the rotor head pocket **28**.

[0024] In use, the canopy apparatus **10** is mounted to the helicopter **70** to provide shade to an area

adjacent to the helicopter **70**. The rotor blade pockets **12** are placed around the free ends **74** of corresponding rotor blades **72** of the helicopter **70** which lie radially adjacent to each other. The rotor head pocket **28** receives the rotor head **76** of the helicopter **70** therein through the opening **30** to mount the rotor head pocket **28** to the rotor head **76**. When the canopy apparatus **10** is mounted to the helicopter **70** each mounting edge **22** of the flexible sheet **16** lies adjacent and parallel to a corresponding rotor blade **72** of the pair of rotor blades **72**. Individuals may find refuge from weather elements such as radiant heat from the sun or precipitation by standing below the canopy apparatus **10**. The shade may help mechanics or other workers perform work on the helicopter **70** while being protected from these weather elements.
[0025] The cylindrical shape of the rotor head pocket **28** facilitates movement of the canopy apparatus **10** to a desired position by rotating the rotor blades **72** with respect to a body **78** of the helicopter **70**. Other shapes for the rotor head pocket **28** are contemplated which are suitable for facilitating this movement.
[0026] The canopy apparatus **10** may be placed in tension between the rotor head **76** and the free ends **74** of the corresponding rotor blades **72** to further secure the canopy apparatus **10** to the helicopter **70**. In embodiments where fasteners having connecting straps of adjustable length couple the flexible sheet **16** to the rotor head pocket **28**, the connecting straps may be cinched as desired to adjust the tension in the canopy apparatus **10**.
[0027] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.
[0028] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A canopy apparatus configured to mount to a helicopter, the apparatus comprising: a pair of rotor blade pockets, each rotor blade pocket being shaped and configured to receive therein a free end of a corresponding rotor blade of a pair of rotor blades of the helicopter which lie radially adjacent to each other; a flexible sheet being coupled to and extending between the pair of rotor blade pockets, the flexible sheet having a top side and a bottom side, each rotor blade pocket being mounted to a corresponding mounting edge of a pair of mounting edges of the flexible sheet, each mounting edge of the flexible sheet extending from the corresponding rotor blade pocket toward a vertex; and a rotor head pocket coupled to the flexible sheet at the vertex, the rotor head pocket having an opening extending therein which faces downwardly with respect to the flexible sheet during normal use, the rotor head pocket being shaped and configured to receive a rotor head of the helicopter therein.
2. The apparatus of claim 1, wherein each mounting edge of the pair of mounting edges being configured to lie adjacent and parallel to a corresponding rotor blade of the pair of rotor blades of the helicopter.

3. The apparatus of claim 1, wherein the flexible sheet has a triangular shape.
4. The apparatus of claim 1, wherein the rotor head pocket is shaped and configured to permit the rotor blades of the helicopter to rotate with respect to a body of the helicopter.
5. The apparatus of claim 4, wherein the rotor head pocket has a cylindrical shape with a central axis that extends orthogonally through the opening.
6. The apparatus of claim 1, further comprising a plurality of mounting straps coupled to the flexible sheet, each mounting strap of the plurality of mounting straps being mounted to the flexible sheet on an associated mounting edge of the pair of mounting edges, each mounting strap of the plurality of mounting straps being sized and configured to surround a corresponding one of the pair of rotor blades.
7. The apparatus of claim 6, wherein each mounting strap comprises a pair of segments and a connector for attaching the pair of segments to each other.
8. The apparatus of claim 7, wherein the connector of each mounting strap comprises a parachute buckle.
9. The apparatus of claim 6, wherein each mounting strap is adjustable in length such that the mounting strap is cinchable.
10. The apparatus of claim 6, wherein the rotor head pocket is sized to receive the rotor blade pockets, the flexible sheet, and the mounting straps therein through the opening to position the canopy apparatus in a stored configuration.
11. The apparatus of claim 1, wherein the rotor head pocket is sized to receive the rotor blade pockets and the flexible sheet therein through the opening to position the canopy apparatus in a stored configuration.
12. A canopy apparatus configured to mount to a helicopter, the apparatus comprising: a pair of rotor blade pockets, each rotor blade pocket being shaped and configured to receive therein a free end of a corresponding rotor blade of a pair of rotor blades of the helicopter which lie radially adjacent to each other; a flexible sheet being coupled to and extending between the pair of rotor blade pockets, the flexible sheet having a top side and a bottom side, each rotor blade pocket being mounted to a corresponding mounting edge of a pair of mounting edges of the flexible sheet, each mounting edge of the flexible sheet extending from the corresponding rotor blade pocket toward a vertex, each mounting edge of the pair of mounting edges being configured to lie adjacent and parallel to a corresponding rotor blade of the pair of rotor blades of the helicopter, the flexible sheet having a triangular shape; a rotor head pocket coupled to the flexible sheet at the vertex, the rotor head pocket having an opening extending therein which faces downwardly with respect to the flexible sheet during normal use, the rotor head pocket being shaped and configured to receive a rotor head of the helicopter therein, the rotor head pocket having a cylindrical shape with a central axis that extends orthogonally through the opening; and a plurality of mounting straps coupled to the flexible sheet, each mounting strap of the plurality of mounting straps being mounted to the flexible sheet on an associated mounting edge of the pair of mounting edges, each mounting strap of the plurality of mounting straps being sized and configured to surround a corresponding one of the pair of rotor blades, each mounting strap comprising a pair of segments and a connector for attaching the pair of segments to each other, each mounting strap being adjustable in length such that the mounting strap is cinchable, the connector of each mounting strap comprising a parachute buckle; wherein the rotor head pocket is sized to receive the rotor blade pockets, the flexible sheet, and the mounting straps therein through the opening to position the canopy apparatus in a stored configuration.